



1.06 Shaling Out

1. Purpose

The purpose of this Standard Operating Procedure (SOP) is to ensure that the personnel involved in entering this confined space are protected from the hazards associated with it, and to ensure that any impact to the environment is minimal. Every person working on this project must read and understand this SOP.

LOW HAZARD ATMOSPHERE REMINDER :

The following procedure is only for Low Hazard Atmospheres only!!

The air inside the vacuum tank must read

0 - LEL

0 - H₂S

0 - CO

20.9 - O₂

If the Monitor reads anything other than clean air, entry into the Vacuum Tank is not permitted for any reason. Contact your Supervisor for further instructions

2. Scope

This SOP applies to every person performing confined space work, supervising the project, confined space safety watches and other individuals performing other duties associated with this confined space entry/project. **This SOP does not address working in Confined Spaces with Inert or Toxic Atmospheres**

3. Hazard Assessment & Identification

Listed below are typical hazards expected in any general confined space entry. It's required that; a complete site-specific hazard assessment must be completed reviewed /signed off by project supervisor or designate before each project starts to identify the specific hazards for the project/site. In British Columbia all hazard assessments must be signed off by a Professional Body.

Examples of some of the hazards are listed below may include but not limited too are as follows .

- Thermal Energy -Combustion /Fire/Explosion
- Exposure To Flammable Substance
- Exposure To Toxic/Carcinogenic Substance
- Exposure To Corrosive Substance



- Exposure To Irritating /Sensitizing Substance
- Temperature Extremes -Personal
- Potential Energy -Pressurized Gases /Liquids
- Potential Energy -Stored Energy
- Deficient Atmosphere -Toxic or Lack of Oxygen
- Physical Stress -Personal Injury
- Physical Stress -Reduced or Restricted Visibility
- Physical Stress -Excessive Noise
- Mental Stress -Confined Space
- Personal Injury Slipping , Tripping Hazard
- Fall from Height – Equipment /Personal
- Ergonomics
- Equipment Failure /Stress
- Mud , Ice, Weather
- Lifting , Twisting , Pushing , Pulling
- Wind Direction

4. EHS Requirements

4.1 Personal Protective Equipment

Confined space entry can potentially expose workers to some very unique hazards that may cause serious injury or death. The appropriate personal protective equipment (PPE) must be worn when working in the project area. PPE provides a barrier between the worker and the material and therefore offers a higher level of protection than working without it. It is, however, not as effective as elimination of the hazard, engineered controls, or administrative controls.

The following PPE is standard PPE requirements for all Silverpoint personal. Refer to Hazard and Risk Assessment for specific PPE for the project.

- Nomex III Coveralls
- Hard Hat
- Steel Toed Boots
- Safety Glasses



- Gloves
- Personal Monitor Measuring All Known or Present Gases .

4.2 Rescue Equipment

Check List and Job Specific Hazard Assessment for Specific Rescue Equipment Required for Project .

- Harnesses
- Lanyards
- Monitors
- First Aid Kit
- Flashlights

4.3 Communication System

Check List and Job Specific Hazard Assessment for Specific Communication Equipment Required for Project .

-Verbal – Hand Signals – Light Signals – Rope Tugging – Knocking or Tapping

4.4 Equipment Inspection

Check List and Job Specific Hazard Assessment for Instructions on Equipment that Needs Inspection .

- Harnesses & Lanyards
- Monitors
- Fire extinguishers
- Eye wash Station
- First Aid Kits

4.5 Atmospheric Testing and Entry Requirements

Check List and job specific hazard assessment for specific testing parameters equipment and methods for required project . Monitor Bump test logs entry requirements Alberta Regulations entry permits .

Bump Testing Procedure for the GX-2009 only.

If you are using a different type of Monitor , follow the Manufactures Procedure .

The following must be performed in a known Clean Atmosphere

- Press the power /mode button for 1 second until it beeps to turn the monitor on.
- The monitor will display the days remaining until a calibration is needed .
- The monitor will go thru its warm up cycle and perform a self test .
- Monitor will perform an automatic fresh air adjustment ; be sure you are in a fresh air.



- Monitor should display fresh air readings , 0-Co, 0-H2s, 0-LEL and 20.9 – 02.
- Place the calibration cup, hose and balloon assembly on the bottom of the monitor .
- Calibration cup only goes on 1 way, look for the 4 small round indents and match them to the monitor .
- Put the small clear tube into the bump gas nozzle .
- Kink the clear tubing before you put bump gas into the balloon .
- Slowly pull bump test gas canister trigger until a small amount of bump gas starts to flow.
- Monitor should alarm (beep, vibrate, flashing red light). All 4 numbers must be flashing .
- Remove calibration cup from monitor to allow fresh air to reach monitor heads .
- Press the power /mode button to go to peak reading screen and record numbers of the bump test.
- To zero peak reading , press the AIR button until it tells you to release .

5. References

- Silverpoint Confined Space Code of Practice
- Alberta OH&S
- Silverpoint Task List
- Silverpoint THA 1.06

6. Procedure for Shaling Out

The truck must be shaled out into Shale Bin/Pit or appropriate burned area as anything inside the tank is considered Drilling Waste

- 6.1. After you finish unloading fluid from the tank you may still have solids on the truck. You now have to shale out. Remember if you have to enter the tank then you need to fill out a Confined Space Entry Permit .
- 6.2. Move truck to corner of pit. Where the pile of shale will be out of the way. This way the backhoe can push the shale pile into the pit. Make sure that you check for wind direction .
- 6.3. Never work on the back of the truck or on the rear deck while it is next to the open pit. If you must open the tank and enter it pull the truck ahead a minimum of 2 meters . Put truck on vacuum to loosen off wing nuts on rear door. Then back up to the pit again . Leave rear door shut so you don't make a mess . **Never** put truck on pressure to open rear door.
- 6.4. When backed up to pit then release the vacuum by moving the 4-way valve to neutral then open rear door slowly . Raise tank up. When tank is almost to the top go slowly, so you don't put stress on ram when you reach the max. Watch the rear door when raising the tank makes sure it won't hit the beaver tail.
- 6.5. When tank is up, start to vibrate solids out. Vibrate for about 30 seconds , and then look inside tank. If needed , try



vibrating again. If you can't get the solids out this way, then you will have to enter tank to shovel it out.

6.6. Lower tank, insert safety bar.

Please remember that before you enter the tank then you must complete a Confined Space Entry Form and follow those procedures .

7. Entering Tank and Removing Shale

- 7.1. Safely back into area where you will be cleaning the tank. Do not clean tank out over a pit.
- 7.2. Operator and safety standby person will start required confined space entry permit .
- 7.3. Inspect and put on all required PPE. Record the results of harness and lanyard inspection on the confined space entry permit .
- 7.4. Bump test the 4-gas monitor and record the results in the bump test section of the confined space entry permit . Always follow instructions in this SOP for bump testing .
- 7.5. Operator and safety standby person will complete all sections of the confined space entry permit prior to entering confined space
- 7.6. Engage Hibon on vacuum , build vacuum to assist in opening rear door wing nuts .
- 7.7. ALWAYS FOLLOW CONFINED SPACE PRODEDURE FOR TANK ENTRY**
- 7.8. While vacuum is building test atmosphere at the vacuum system exhaust port . Atmosphere test must show that tank is a low hazard atmosphere .
- 7.9. Record the results on the confined space entry permit .
- 7.10. Undo rear door wing nuts .
- 7.11. Release vacuum and open rear door and install rear door safety bar, slowly lower rear door onto the safety bar.
- 7.12. Switch Hibon pump to pressure and ventilate the tank for 2 minutes before entering tank .
- 7.13. Ventilation and monitoring of the vacuum tank must be continuous while working inside the tank .
- 7.14. Safely enter tank using 3 points of contact and start cleaning and / or inspection .
- 7.15. Once tank cleaning is complete clean the rear door seal and tank lip.
- 7.16. Safely exit tank using 3 points of contact .
- 7.17. Raise rear door and remove safety bar.
- 7.18. Slowly lower the rear door and close rear door wing nuts .



- 7.19. Switch Hibon pump to vacuum to assist in closing rear door.
- 7.20. Tighten rear door wing nuts.
- 7.21. Shut down hibon and PTO.
- 7.22. Operator and safety standby person will sign off that confined space is complete .

Remember : if you are in doubt about what to do, call your Supervisor